



FINAL MARK

GIRRAWEEN HIGH SCHOOL
Mathematics Extension 1
HSC ASSESSMENT
Task 2 – 2010
Half yearly examination
ANSWERS COVER SHEET

Name: _____ Teacher: _____.

QUESTION	MARK	E2	E3	E4	E5	E6	E7
1	/16						✓
2	/15						✓
3	/16						✓
4a	/5	✓					✓
b	/4		✓				✓
c	/6						✓
	/15						
5	/15						✓
6	/23						✓
TOTAL	/100	/4	/9				/100

HSC Outcomes

Mathematics Extension 1

- HE1 appreciates interrelationships between ideas drawn from different areas of mathematics.
- HE2 uses inductive reasoning in the construction of proofs.
- HE3 uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion and exponential growth and decay.
- HE4 uses the relationship between functions, inverse functions and their derivatives
- HE5 applies the chain rule to problems including those involving velocity and acceleration as functions of displacement.
- HE6 determines integrals by reduction to a standard form through a given substitution.
- HE7 evaluates mathematical solutions to problems and communicates them in an appropriate form.



GIRRAWEE HIGH SCHOOL
HALF YEARLY EXAMINATION

2010

MATHEMATICS
EXTENSION 1

Time allowed - Two hours
(Plus 5 minutes' reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

Total Marks – 100

Attempt all questions 1-6

All questions are NOT of equal value.

Answer each question clearly ON A SEPARATE PAGE!

Question 1 (16 Marks) Use a separate piece of paper.

Marks

(a) Solve for x : $\frac{4}{5-x} \geq 1$

5

(b) The point $P(-3, 8)$ divides the interval AB externally in the ratio $k : 1$. If A is the point $(6, -4)$ and B is the point $(0, 4)$, find the value of k .

3

(c) The acute angle between the lines $y = 3x + 5$ and

$y = mx + 4$ is 45° . Find the two possible values of m .

3

(d) The polynomial $P(x) = x^3 + ax + 12$ has a factor $(x + 3)$. Find the value of a .

1

(e)

(i) Find the domain of the function $y = \ln(x - 3)$

1

(ii) Sketch the graph of $y = \ln(x - 3)$, showing any asymptotes and any intercepts on the co-ordinate axis.

3

Question 2 (15 Marks) Use a separate piece of paper.

(a) Differentiate:

i) $\frac{\log_e x}{x}$ 2

ii) $x^3 e^{-2x}$ 2

iii) 5^{3x-2} 3

iv) $\log_e \left[\frac{x+4}{x-3} \right]$ 3

(b) Find the equation of the normal to the curve $y = \ln x$ at the point where $x = 1$ 4

Question 3 (16 Marks) Use a separate piece of paper.

(a) Evaluate i) $\int_0^2 e^{5-2x} dx$ 3

ii) $\int_1^3 \frac{x}{x^2+1} dx$ 3

(b) Find i) $\int \frac{x+1}{\sqrt{x}} dx$ 2

ii) $\int \left(1 + \frac{3}{x-2} \right) dx$ 3

(c) (i) Find $\frac{d}{dx} e^{3x^2}$ 2

(ii) Hence find $\int x e^{3x^2} dx$ 3

Question 4 (15 Marks) Use a separate piece of paper.

(a) Use mathematical induction to prove that, for every positive integer n ,

$$13 \times 6^n + 2 \text{ is divisible by } 5$$

5

(b)

(i) Explain why the probability of obtaining 2 heads and a tail when three

1

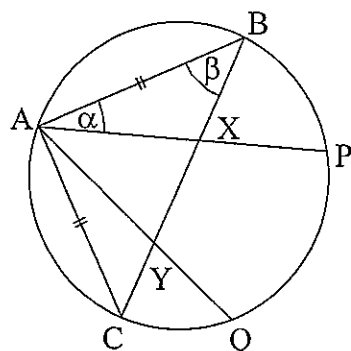
coins are tossed is $\frac{3}{8}$.

(ii) Sian tosses three coins 10 times in a row. Calculate the probability of obtaining 2 heads and a tail at least 2 times. Give your answer correct to 3 significant figures.

3

(c)

6

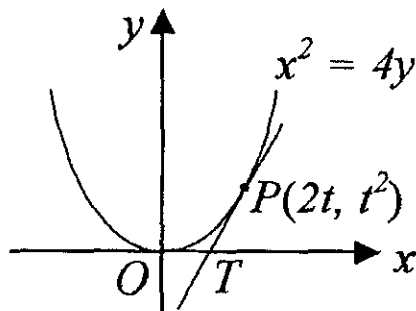


Let ABPQC be a circle such that $AB = AC$, AP meets BC at X , and AQ meets BC at Y , as in the diagram. Let $\angle BAP = \alpha$ and $\angle ABC = \beta$.

- i. Copy the diagram into your Writing Booklet and state why $\angle AXC = \alpha + \beta$.
- ii. Prove that $\angle BQP = \alpha$.
- iii. Prove that $\angle BQA = \beta$.
- iv. Prove that PQYX is a cyclic quadrilateral.

Question 5 (15 Marks) Use a separate piece of paper.

a)



$P(2t, t^2)$ is a variable point which moves on the parabola $x^2 = 4y$.
The tangent to the parabola at P cuts the x axis at T. M is the midpoint of PT

i. Show that the tangent PT has equation $tx - y - t^2 = 0$ 2

ii. Show that M has coordinates $(\frac{3t}{2}, \frac{t^2}{2})$. 2

iii. Hence find the Cartesian equation of the locus of M as P moves on the parabola. 4

(b) Find the value of the term that is independent of x in the expansion

of $\left(x^2 + \frac{3}{x}\right)^6$ 3

(c) The area enclosed between the curves $y = e^x$, $y = e^{\frac{1}{2}x}$ and the line $x = 2$ is rotated about the x axis. Find the volume of the solid generated. 4

Question 6 (23 Marks) Use a separate piece of paper.

(a)

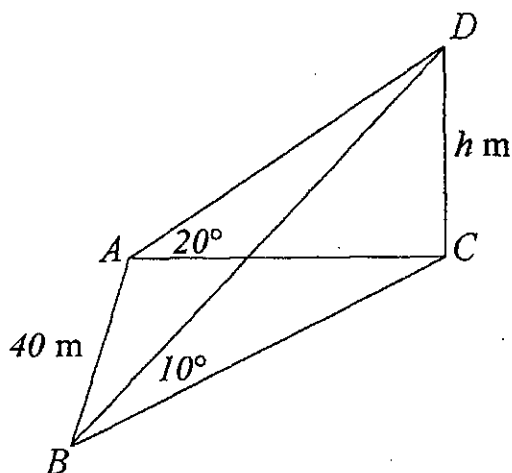


FIGURE NOT TO SCALE

A vertical flagpole CD of height h metres stands with its base C on horizontal ground. A is a point on the ground due West of C and B is a point on the ground 40 metres due South of A. From A and B the angles of elevation of the top D of the flagpole are 20° and 10° respectively. Find the height of the flagpole correct to the nearest metre.

6

(b) (i) Factorize $a^3 + b^3$.

1

(ii). Hence, or otherwise, show that $\frac{2 \sin^3 A + 2 \cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$,

if $\sin A + \cos A \neq 0$

4

(c) Solve $7 \sin x - 4 \cos x = 4$, for $0^\circ \leq x \leq 360^\circ$, by using the t formulae

6

(d) (i) Express $\cos x - \sin x$ in the form $R \cos(x + \alpha)$

where α is in degrees.

3

(ii) Hence solve the equation $\cos x - \sin x = -1$

for $0^\circ \leq x \leq 360^\circ$

3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



Question 1

(a) By critical points: (5)

Equality $\frac{4}{5-x} = 1$

$4 = 5-x$

$x = 1$

Discontinuity: $x \neq 5$

Testing.



$x=6, \frac{4}{5-6} = -4 > 1$

$\frac{4}{5-3} = 2 > 1$

$\frac{4}{5-0} = \frac{4}{5} < 1$

False

True

$1 \leq x < 5$

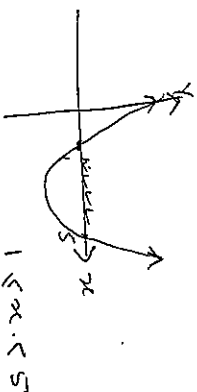
OR

$\frac{(5-x)^2 \times 4}{5-x} > (5-x)^2; x \neq 5.$

$4(5-x) > (5-x)^2$

$(5-x)^2 - 4(5-x) \leq 0$

$(5-x)(5-x-4) \leq 0 \Rightarrow (5-x)(1-x) \leq 0$



b) $A(6,4), B(0,4)$ (3)

$K, -1, P(-3,8)$

$-3 = \frac{-1 \times 6 + K \times 0}{K-1} = \frac{-6}{K-1}$

$\therefore -2K + 3 = -6 \Rightarrow 3K = 9$
 $\therefore K = 3$

c) $y = 3x + 5 \Rightarrow m_1 = 3$

$y = mx + 4 \Rightarrow m_2 = m$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$\tan 45^\circ = \left| \frac{3 - m}{1 + 3m} \right| = 1$

$\therefore \frac{3-m}{1+3m} = 1 \text{ or } -\frac{(3-m)}{1+3m} = 1$

$\Rightarrow 3-m = 1+3m \text{ or } -3+m = 1+3m$

$\therefore m = \frac{1}{2}$

(3)

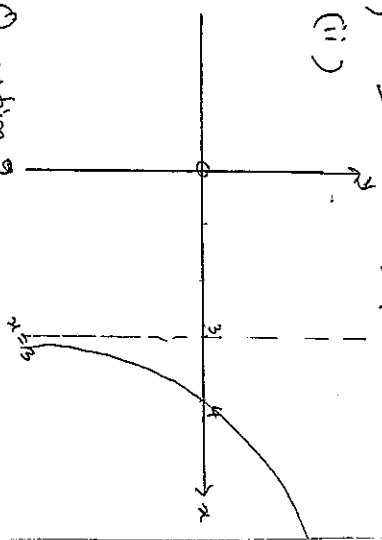
d) $R(x) = x^3 + ax + 12$ (2)

$P(-3) = -3^3 + 3a + 12 = 0$

$\Rightarrow -27 + 3a + 12 = 0 \Rightarrow 3a = 15$
 $\therefore a = 5$

Domain: $x > 3$

(ii)



Question 2

g) (i) $y = \log_e x$

$\therefore \frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2}$

$= \frac{1 - \log_e x}{x^2}$ (2)

(ii)

$y = x^3 e^{-2x}$

$\frac{dy}{dx} = 3x^2 \cdot e^{-2x} + (e^{-2x} \cdot -2)x^3$
 $= x^2 e^{-2x} (3 - 2x)$ (2)

(iii)

$y = 5^{3x-2}$

$y' = 5^{3x-2} \cdot \log 5 \cdot 3$

$= 3 \log 5 \cdot 5^{3x-2}$ (3)

(iv)

$y = \log_e \left(\frac{x+4}{x-3} \right)$

$= \log_e (x+4) - \log_e (x-3)$

$\therefore y' = \frac{1}{x+4} - \frac{1}{x-3}$ (3)

b) $y = \ln x$

$y' = \frac{1}{x}$

at $x=1, y' = \frac{1}{1} = 1$

$\therefore y_{\text{normal}} = -1$
at $x=1, y = \ln 1 = 0$

$\therefore$ equation is

$y-0 = -1(x-1) = -x+1$

$y+x-1=0$ (4)

Question 5

(b) $T_{r+1} = b C_r (x^2)^{6-r} \cdot \left(\frac{3}{x}\right)^r$

$= b C_r \cdot x^{12-2r} \cdot 3^r \cdot x^{-r}$

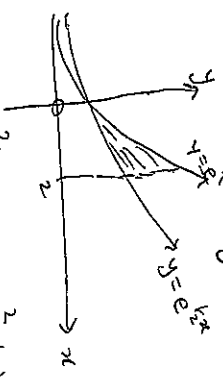
$= b C_r \cdot 3^r \cdot x^{12-3r}$

term independent of x

$\Rightarrow 12-3r=0 \Rightarrow r=4$

$\therefore T_5 = b C_4 \cdot 3^4 = 1215$ (3)

(c) $V = \pi \int y^2 dx$



$V = \pi \int_0^2 (e^{kx})^2 dx - \int_0^2 (e^{kx})^2 dx$

$= \pi \int_0^2 (e^{2kx} - e^{kx}) dx$

$= \pi \left[\frac{e^{2kx}}{2} - e^{kx} \right]_0^2$

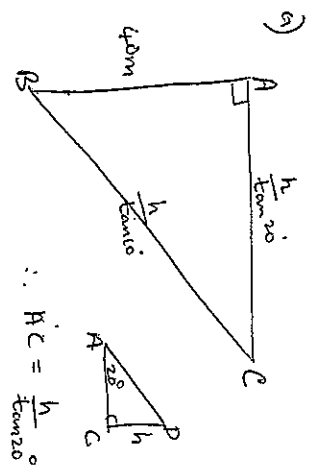
$= \pi \left[\frac{e^4}{2} - e^2 - \left(\frac{1}{2} - 1 \right) \right]$

$= \pi \left[\frac{e^4}{2} - e^2 + \frac{1}{2} \right]$

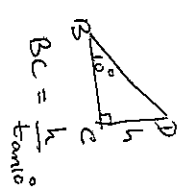
$= \frac{\pi}{2} [e^4 - 2e^2 + 1]$

(4)

Question 6



$\therefore AC = \frac{h}{\tan 20^\circ}$



$\therefore \left(\frac{h}{\tan 10^\circ} \right)^2 - \left(\frac{h}{\tan 20^\circ} \right)^2 = 40^2$

$h^2 \left[\left(\frac{1}{\tan 10^\circ} \right)^2 - \left(\frac{1}{\tan 20^\circ} \right)^2 \right] = 40^2$

$\therefore h^2 = 65$
 $h \approx 8m$

(6)

b) (i) $a^2 + b^2 = (a+b)(a^2 - ab + b^2)$ (1)

(ii) $\frac{2 \sin^3 A + 2 \cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$

LHS

$\frac{2(\sin^3 A + \cos^3 A)}{\sin A + \cos A} =$

$= \frac{2(\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A)}{\sin A + \cos A}$

$= 2(1 - \sin 2A) = 2 - \sin 2A$

Question 7

$\tan \frac{\alpha}{2} = t$

(c)

$T \sin \alpha - 4 \cos \alpha = 4$

$\frac{T \cdot 2t}{1+t^2} - \frac{4(1-t^2)}{1+t^2} = 4$

$14t - 4 + 4t^2 = 4 + 4t^2$

$\therefore 14t = 8$

$t = \frac{4}{7}$

$\tan\left(\frac{\alpha}{2}\right) = \frac{4}{7}$

$\therefore \frac{\alpha}{2} = \tan^{-1} \frac{4}{7}$

$\therefore \alpha = 2 \tan^{-1} \frac{4}{7}$

Check $\theta = 180^\circ$ in $7 \sin \alpha - 4 \cos \alpha = 4$

LHS $7 \times 0 - 4 \times -1 = 4 = RHS$

$\therefore \alpha = 180^\circ$ is also a solution.

$\therefore \alpha = 180^\circ$ or $\alpha = 2 \tan^{-1} \frac{4}{7}$ (6)

(d) (i) $\cos \alpha - \sin \alpha = R \cos(\alpha + \phi)$

Expanding,

$\cos \alpha - \sin \alpha = R \cos \alpha \cos \phi - R \sin \alpha \sin \phi$

$\therefore R = \sqrt{1+1} = \sqrt{2}$

As $\cos \alpha = R \cos \alpha \cos \phi$,

$\cos \alpha = \frac{1}{R} = \frac{1}{\sqrt{2}}$

As $\sin \alpha = R \sin \alpha \sin \phi$,

$\sin \alpha = \frac{1}{\sqrt{2}}$

$\therefore \alpha = 45^\circ$ or 135°

(ii) $\sqrt{2} \cos\left(\alpha + 45^\circ\right) = -1$ (2)

$\therefore \cos\left(\alpha + 45^\circ\right) = -\frac{1}{\sqrt{2}}$

$\therefore$ related angles are 135° or 225°

$\therefore \alpha + 45^\circ = 135^\circ$ or $\alpha + 45^\circ = 225^\circ$

$\therefore \alpha = 90^\circ$ or $\alpha = 180^\circ$ (4)